

Executive Summary

Rwanda NBSAP Monitoring Dashboard 2025–2030

Prepared for: Rwanda Environment Management Authority (REMA) **System:** <https://nbsap-dashboard-rw.vercel.app> **Date:** June 2026 **Version:** 1.0

1. Purpose and Context

Rwanda is a signatory to the **Convention on Biological Diversity (CBD)** and has committed to the **Kunming-Montreal Global Biodiversity Framework (KM-GBF)** adopted in December 2022. Under this commitment, Rwanda developed its **National Biodiversity Strategy and Action Plan (NBSAP) 2025–2030**, which defines **22 national biodiversity targets** aligned to the 23 global KM-GBF targets across 4 strategic goals.

The **NBSAP Monitoring Dashboard** is a purpose-built national digital platform that enables Rwanda to:

- **Track** real-time progress against all 22 NBSAP targets across 30 districts
- **Report** biodiversity data through 7 structured reporting tools covering every NBSAP sector
- **Verify** all incoming data through a mandatory quality-assurance workflow before it affects national metrics
- **Visualise** geographic coverage, indicator trends, and compliance status
- **Export** data in formats suitable for CBD Secretariat reporting obligations

The system replaces the previous fragmented approach of manual spreadsheets, PDF reports, and email-based review cycles with a single, authoritative, real-time source of truth for Rwanda's biodiversity data.

2. System Capabilities

2.1 Real-Time National Progress Tracking

The dashboard provides an executive-level view of Rwanda's biodiversity status:

Capability	Description
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22 National Targets	Full NBSAP 2025–2030 target set with progress bars, status indicators, baselines, and implementation milestones
4-Tier Indicator Hierarchy	~80 indicators across Headline, Component, Binary, and Complementary tiers following KM-GBF structure
Automated Metrics	13 system-wide metrics auto-calculated from approved reports (forest/wetland restoration, finance, HWC incidents, EIA compliance)
3-Colour Progress System	Consistent green (>=70%), amber (40–69%), red (<40%) status across every page

2.2 Multi-Stakeholder Reporting (7 Tools)

Tool	Domain	Key Data Collected
T01	National Institutional Reporting	Institution, budget utilised, compliance score, policy progress
T02	Ecosystem & Habitat Monitoring	District, forest hectares, wetland hectares, land cover change
T03	Protected Area Management	Area name, coverage, management effectiveness, illegal cases, restoration
T04	Human-Wildlife Conflict	Incident type, district, species, HWC case count, mitigation measures
T05	Finance & Resource Mobilisation	Budget allocated, budget disbursed, finance source
T06	Private Sector EIA Compliance	Company, EIA status, sector, restoration commitments
T07	Community Engagement	Community name, district, participant count, traditional knowledge

2.3 Data Quality Assurance

All submitted data passes through a **mandatory Verification Queue** before affecting national metrics:

1. Reporter submits via structured form with evidence uploads
2. Submission enters queue with **pending** status
3. Authorised reviewer examines data and evidence
4. Reviewer approves (metrics auto-update) or rejects (submitter notified with reason)
5. Every decision is logged in the immutable audit trail

2.4 Role-Based Access Control

The system enforces **6 distinct user roles** with carefully scoped permissions:

Role	Users	Access Level
REMA Administrator	REMA national team	Full system access: verification, user management, audit log, all exports
Lead Government Ministry Reporter	Ministry focal points	Submit reports, verify queue, analytics, risk register
District Reporting Officer	30 district officers	Submit reports for own district
Policy Monitor	National policymakers	Read-only strategic dashboards, analytics, risk register
Development Partner	NGOs, research, private sector	Analytical access for programme alignment
Public Viewer	General public	View-only: dashboard, targets, indicators, map

2.5 Geographic Coverage

- **30 districts** across 5 provinces with interactive map visualisation
- **GeoJSON layers** for protected areas, river networks, lakes
- **District-level** compliance tracking and reporting status
- **GBIF integration** for species occurrence data
- **RBIS integration** for Rwanda Biodiversity Information System access

3. Technology Summary

3.1 Architecture

The system follows a **serverless Backend-as-a-Service (BaaS)** architecture:

Layer	Technology	Hosting
Frontend	React 18 + TypeScript + Vite	Vercel (global CDN)
Backend	Supabase (PostgreSQL + Auth + Storage + Realtime)	Supabase Cloud
Database	PostgreSQL with Row-Level Security	Supabase managed
File Storage	Supabase Storage (S3-compatible)	Supabase managed
Real-time	WebSocket (Supabase Realtime)	Supabase managed
Geo Data	Static GeoJSON + GBIF API proxy	Vercel + Supabase Edge Functions

3.2 Key Technical Features

Feature	Implementation
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Automated metric calculation	PostgreSQL triggers fire on report approval — no manual calculations required
Real-time dashboard updates	WebSocket push notifications to all connected browsers
Delete reversal	When an approved report is deleted, all metric effects are automatically reversed
Weighted progress	Each tool has a configurable weight affecting its contribution to target progress
Profile caching	Two-layer cache (memory + sessionStorage) with 5-minute TTL for instant page loads
Code splitting	Every page lazy-loaded via React.lazy() — initial bundle is minimal
Immutable asset caching	Content-hashed filenames with 1-year cache headers for instant repeat loads
Stale chunk recovery	Auto-reload when deployment invalidates old chunk URLs

3.3 Security Controls

Control	Implementation
Authentication	JWT tokens via Supabase Auth with auto-refresh
Row-Level Security	25+ PostgreSQL RLS policies — each user only sees authorised data
Transport	HTTPS/TLS on all traffic
Security Headers	CSP, X-Frame-Options: DENY, X-Content-Type-Options: nosniff, X-XSS-Protection
Account Controls	Admin suspension/deactivation with auto-reactivation on expiry
Audit Trail	Immutable append-only log of every system action
Verification Gate	No data affects national metrics until approved

4. Database Overview

4.1 Schema Summary

Category	Tables	Purpose
User Management	profiles, user_settings, notification_preferences, role_change_requests	User accounts, preferences, role workflows
Core Data	nbsap_targets, indicators, nbsap_milestones	22 targets, ~80 indicators, 63 milestones

Reporting	toolkit_reports, report_attachments	T01–T07 submissions with evidence files
Metrics	system_metrics, tool_weights	13 auto-calculated metrics, 7 tool weight configs
Governance	compliance_records, risks, audit_log	Compliance tracking, risk register, audit trail
Notifications	notifications	In-app notification delivery
Geography	districts, provinces, lakes	30 districts, 5 provinces, inland water bodies

4.2 Key Automation

- **7 database triggers** handle automated metric updates, notification delivery, and role change workflows
- **10+ database functions** encapsulate business logic (weighted progress, metric recalculation, role checks)
- **20+ database indexes** ensure fast queries on frequently filtered columns
- **Full RLS** on all 18 tables with role-aware policies

5. NBSAP Target Coverage

The system tracks all 22 Rwanda NBSAP 2025–2030 targets organised by KM-GBF Goal:

Goal	Targets	Focus
Goal A (Targets 1–4)	Reduce threats to biodiversity	Spatial planning, ecosystem restoration, protected areas, species conservation
Goal B (Targets 5–8)	Meet people's needs	Sustainable harvesting, invasive species, pollution, climate resilience
Goal C (Targets 9–12)	Tools and solutions	Community benefits, sustainable agriculture, ecosystem services, urban green spaces
Goal D (Targets 13–22)	Means of implementation	ABS, mainstreaming, business disclosure, finance, capacity, data access, inclusion

Each target includes:

- Official Rwanda-specific target statement and baseline data
- Phase-by-phase implementation milestones (2025–2030)
- Linked headline and component indicators
- Responsible stakeholders
- Progress tracking via tool-weighted automated calculations

6. Reporting & Compliance

6.1 Export Capabilities

Format	Purpose
CSV	Raw data export for spreadsheet analysis
PDF	Formatted progress reports for printing and distribution
JSON	Structured data export/import for system interoperability

6.2 CBD Reporting Alignment

The indicator framework follows the official KM-GBF three-tier structure:

- **Tier 1 — Headline Indicators:** High-level, internationally comparable metrics required for CBD national reporting
- **Tier 2 — Component Indicators:** Detailed implementation metrics supporting adaptive management
- **Tier 3 — Binary Indicators:** Yes/No policy condition assessments

Data collected via the 7 reporting tools maps directly to the indicator hierarchy, enabling direct export for Rwanda's periodic National Reports to the CBD Secretariat.

7. Deployment & Operations

Aspect	Details
Live URL	https://nbsap-dashboard-rw.vercel.app
Frontend hosting	Vercel (auto-deploy on git push to main)
Backend hosting	Supabase Cloud (managed PostgreSQL, Auth, Storage)
CI/CD	Git push → Vercel auto-build → global CDN deployment
Uptime target	99.5% (Vercel + Supabase SLAs)
Estimated annual cost	Low hundreds of USD (free-tier eligible at current scale)
Scalability	Designed for 500 concurrent users and 10,000+ reports

8. Migration History

The database has evolved through **22 migrations** (001–022), covering:

- Initial schema with 15 core tables, RLS, and triggers (001)
 - Full Rwanda NBSAP 2025–2030 target data and indicator framework (002, 020)
 - Geographic coordinates for all 30 districts (003)
 - Role change approval workflow with admin constraints (006)
 - Account suspension/deactivation system (009)
 - Public viewer role expansion (010)
 - System metrics automation with 13 metric types (015)
 - Tool weight configuration for progress calculations (016, 017)
 - NBSAP milestone tracking (020)
 - Target baseline and timeline data (021)
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9. Key Differentiators

vs. Manual Approach	NBSAP Dashboard
Weeks of data compilation	Real-time automated metrics
No verification workflow	Mandatory approval queue
Scattered spreadsheets	Single database with RLS
No audit trail	Complete, immutable action log
Role confusion	6-tier enforced access control
Quarterly visibility at best	Live WebSocket updates
Error-prone manual calculations	Database trigger automation
Email-based review	Structured verification queue with evidence

10. Conclusion

The Rwanda NBSAP Monitoring Dashboard provides a production-ready, secure, and scalable national platform for tracking Rwanda's 22 biodiversity targets through 2030. It eliminates manual data compilation, enforces data quality through a verification workflow, automates metric calculations through database triggers, and provides real-time visibility to all stakeholders — from district field officers to Cabinet-level decision-makers to the CBD Secretariat.

The system is live, operational, and processing real data at <https://nbsap-dashboard-rw.vercel.app>.

Prepared by: NBSAP Dashboard Development Team System: Rwanda NBSAP Monitoring Dashboard v1.0.0